

# 1 Separate/Rebuild Serial Numbers

## Usage

This function enables material components listed by serial numbers and already mounted in a component to be exchanged specifically. Mounted material components are demounted and the new material components to be used are mounted. Mounting and demounting are documented in the system and recorded for the relevant serial number.

## Procedure of separating/rebuilding serial numbers

In general, there might be two different situations requiring the exchange of components. At the moment, the system does not differentiate between them.

1. Components are demounted from a part while the process is running  
An operation to "merge" serial number is running and it turns out that a serial number with a "damaged part" has been mounted. The serial number is recorded for an output batch/merged batch "as usual" and is ready to be demounted/rebuilt (batch status = free, class = yield, quality status = free/open).
2. Components are demounted in a subsequent or separate process step (e.g. rework)  
The serial number was recorded for an output batch/merged batch and is ready to be demounted/rebuilt.  
Specific identifiers (customer-specific) specify when a serial number gets to this process (e.g. batch status = blocked, class = rework/scrap, quality status = blocked). But by default these indicators are set: (batch status = free, class = yield, quality status = free/open).

In both cases, the function does not require operations or input components to be logged on. The dialog can be carried out at an independent workstation.

## Posting of serial numbers

The below-mentioned postings have to be performed for the affected components listed by serial numbers, once they have been rebuilt:

### **Serial number of the complete, finished component part**

This serial number is still assigned to the merged batch from which it was removed.

All relevant data (material number, material type, batch status, class, quality status) remains unchanged.

The connection to the demounted component is deleted. Now it is only included in the history. The connection to the newly mounted part is added accordingly.

Serial number data (attributes, document links) can be entered for the rebuilt part.

### Serial number of the demounted component part

This serial number is no longer connected to the entire part from which it was demounted.

For traceability purposes, an entry is made in the batch history (indicating that this component was once mounted in this part).

The serial number of the removed component is not assigned to a new merged batch. It remains as single part (batch with quantity 1) on stock (class = scrap and batch status = blocked).

The merged batch which the demounted serial number derives from also remains unchanged (the inventory is not increased for this batch).

### Serial number of the mounted component part

This serial number is taken from a provided merged batch the inventory of which is reduced accordingly.

As in this process no operation is logged on, no merged batch will be logged on as input batch. The inventory of the merged batch is reduced in the background after mounting.

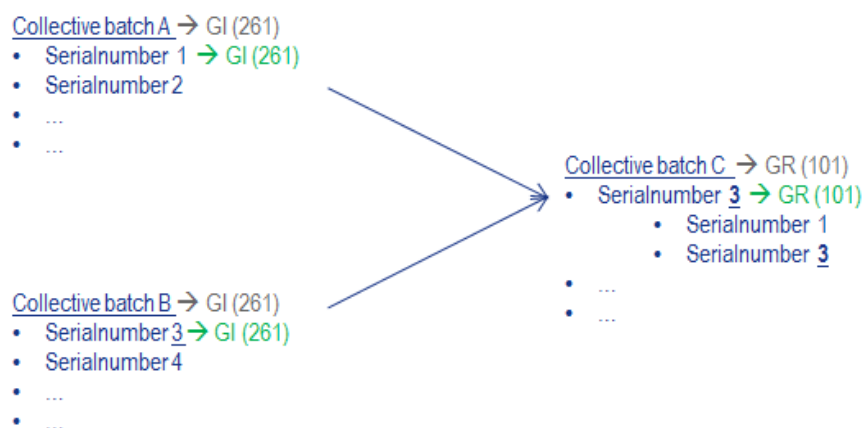
Batch data of the mounted component part is not changed. All relevant data (material number, material type, batch status, class, quality status) remains unchanged.

A connection to the consumed serial numbers/merged batches is established for traceability purposes.

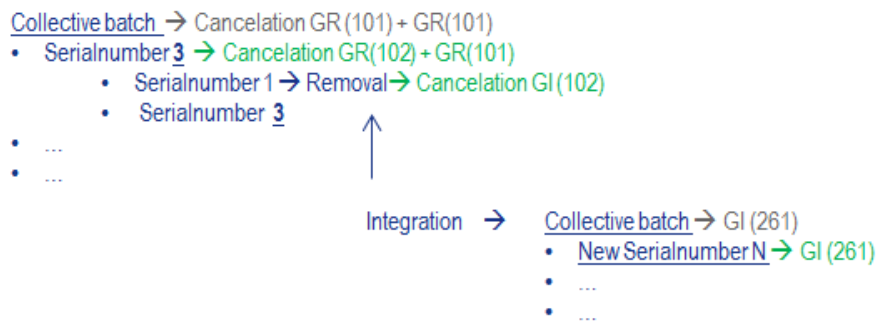
### Generation of goods movements

These goods movements are performed for the mounted/demounted components, the complete component part and for the corresponding merged batches.

Goods movements when merging components listed by serial numbers:



Goods movements when separating components listed by serial numbers:



Subject to configuration, the generation of goods movements can be controlled individually for consumptions as well as for the output material on the level of serial numbers and merged batches. Consequently, it is possible to adjust the upload of these goods movements to the conditions of the relevant ERP system.